Tutorial - 8. 24/03/2026.

Decision Tree.

Email ID	Suspicious Words	Money Keywords	Sender Known.	Has Attachment	Class.
1	8	1	0	1	S
2	1	0	1	0	N.
3.	6	1	0	1	S
4	2	0	1	0	N.
5.	7	1	0	0	S
6	1	0	1	1	N.
7.	9	1	0	1	S.
8	2	0	1	0	N
9	5	1	1	0	S
10.	1	0	1	1	N.

Questions:

1. Which attribute should be selected as the root node? Justify your answer with calculations.
2. Build the complete Decision Tree using the ID_3 algorithm (use Information Gain). Draw the tree structure with all nodes, branches, and leaf nodes clearly labeled.
3. What is the depth of your decision tree? How many leaf nodes does it have?

ID3 -> Entropy & Information Gain.

(IG) -> Information Gain of each feature.

IG -> max -> best feature.

Goal : Reduce Entropy.

↳ Increase Information Gain.

$$\boxed{\text{Entropy} \rightarrow -\sum_{i=1}^{N} P_i * \log_2(P_i)}$$

$$\log_2 P_i = \frac{\log_{10} P_i}{\log_{10} 2} = \frac{\ln P_i}{\ln 2}.$$

IG = Entropy (Parent).

$$\boxed{IG = -\sum_{K} \frac{N_K}{N} \text{Entropy}(K).}$$

① Entropy (Parent or Y) :

$$P_{spam} = \frac{5}{10} = 0.5.$$

$$P_{Nspam} = \frac{5}{10} = 0.5$$

$$\text{Entropy} = -[0.5 * \log_2(0.5) + 0.5 * \log_2(0.5)].$$

$$= 1$$

② Entropy (Money keywords).

1 → spam.
  → non spam(X)

} Perfectly classified so Entropy = 0.

0 → non-spam.
  → spam (X)

$IG(m.k) = 1 - 0 = 1$

③ Entropy (sender known)

1 → spam(1)
  → non spam(5)

0 → spam(4)
  → non spam(0).

for '1'

$$\text{Entropy} = -\left[\frac{1}{6}\log_2\left(\frac{1}{6}\right) + \frac{5}{6}\log_2\left(\frac{5}{6}\right)\right.$$

$$\text{for '0'} \rightarrow 0 + \left.\frac{4}{4}\log_2\left(\frac{4}{4}\right)\right]$$

$= 0.650$

$\simeq 0.65.$

$$IG(S.K) = 1 - \left(\frac{6}{10} \times 0.65 \times \frac{4}{10} \times 0\right)$$

$= 0.61.$

④ Entropy (Has Attachments):-

0 — spam = 2, non-spam = 3 } Entropy > 0.

1 — spam = 3, nonspam = 2 } Entropy > 0.

⑤ Entropy (suspicious words):-

Entropy > 0 (Random value in dataset).

$\boxed{IG < 1}$

∴ Money keywords : Root Node.

If = 1 → spam ; If = 0 → non-spam

branches.

(leaf nodes)

DEPTH = 1 (only one decision node is there)

leaf nodes = 2